Chapter 3: Project Planning

This chapter covers different topics of project planning and some suggestions and ideas from students' perspective. This should be treated as "food for thought", not a step-by-step instruction to perform such planning. Particularly, the chapter discuss about scope planning (Section 3.2), time planning (Section 3.3), risk planning (Section 3.4), communication planning (Section 3.5), way of working (Section 3.6), and usage of tools and infrastructures (Section 3.7). The chapter is particularly written for students who is about or under a software development project.

3.1. Introduction to Project Planning

The project management plan defines how the project is executed, monitored and controlled, and closed. The project management plan's content varies depending on the application area and complexity of the project. Knowledge about project management plan is well presented in books about project management, such as PMBOOK (PMBOOK 2013). In the scope of this book, we do not aim at providing details of every project management knowledge area. The content in this section should be seen as tips to solve most common challenges when students conduct their project management. As described in Figure 5, seven topics about project planning will be presented.

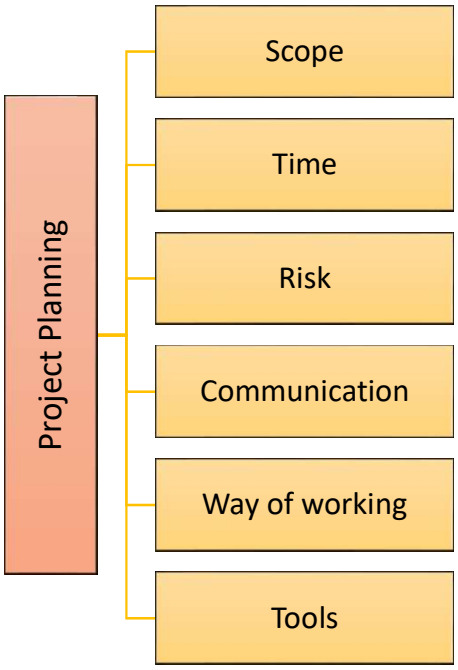

Figure 5: Topics of Student Project Planning

Project planning is widely thought to be an important contributor to project success. Man benefits of project planning has been proof in different project contexts, including clarification

of project uncertainty, reduction of project risks, keeping track of deadlines, delivery of the agreed products. To teach the students about the importance of project planning might not be a simple task. In the beginning, students would probably do not take the lessons through theory lectures. The importance of planning will be better perceived when the students make some mistakes during their projects.

Planning is an iterative effort. Project planning can only be based on available information. At the beginning of a project, there are many uncertainties in the available information and some information about the project and the product may not be available. As the project develops, more and more information become available and uncertainties are resolved. The project plan therefore must be reviewed and updated regularly to reflect this changing information environment. The subsections below present important notes for planning a student project work.

Tip 11: Students should do planning early, accepting imperfection and updating their plan iteratively

3.2 Scope planning

Scope is the defined features and functions of a product, or the scope of work needed to finish a project. There are two ways to look at a project scope:

- Artifact-based scope: for instance, a product composes of three modules. Once the three modules are delivered, the scope of the project is fully covered
- Task-based scope: for instance, a project includes three activities that need to be complete.

Correspondingly, there are two ways of defining a work breakdown structure for a project. A work breakdown structure (WBS)[1] is defined as a hierarchical decomposition of the total scope of work to be carried out by the project team to accomplish the project objectives and create the required deliverables (PMBOOK 2013). WBS provides the basis for planning and managing project schedules, costs, resources, and changes (Schwalbe, 2019, p.213) There are several approaches for developing a WBS, including using guidelines, the analogy approach, the top-down approach, the bottom-up approach, and mind mapping (Schwalbe, 2019, p.233).

[1] https://en.wikipedia.org/wiki/Work_breakdown_structure

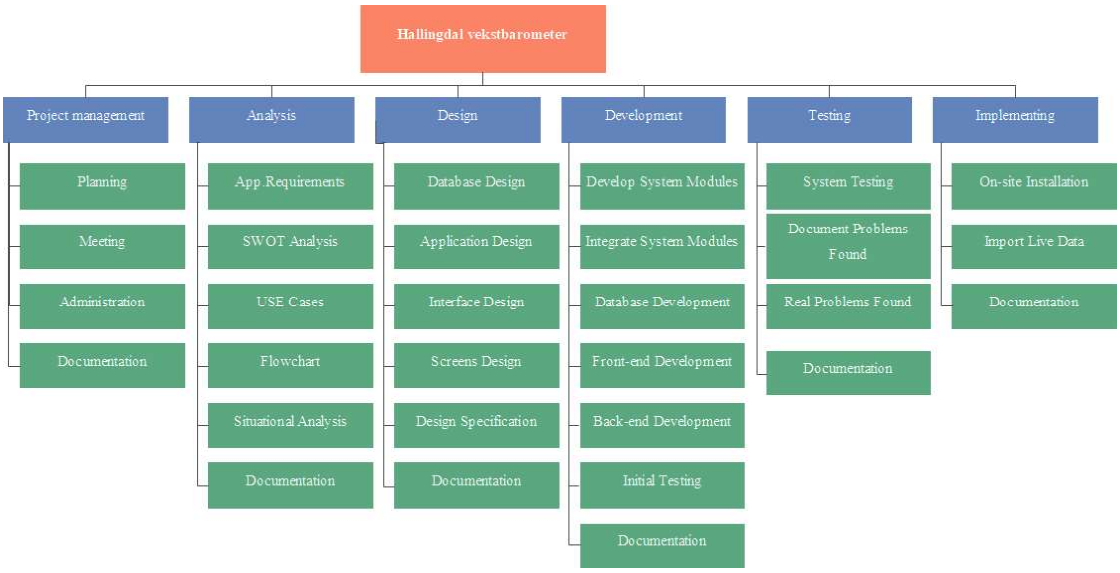

Figure 7: An example of WBS for a software development project

Not every breakdown of project deliverable can be classified as a WBS. WBS has certain characteristics:

- Hierarchy: The WBS is hierarchical in nature. Each "child" level exists in a strict hierarchical relationship with the parent level. The sum of all the child elements should give you the parent element.
- 100% rule: Every level of decomposition must make up 100% of the parent level. It should also have at least two child elements.
- Mutually exclusive: All elements at a particular level in a WBS must be mutually exclusive. There must be no overlap in either their deliverables or their work. This is meant to reduce miscommunication and duplicate work.
- Outcome-focused: The WBS must focus on the result of work, i.e. deliverables, rather than the activities necessary to get there. Every element should be described via nouns, not verbs. This is a big source of confusion for beginners to WBS.

During many years of teaching the course, we observe common mistakes when students learn to make WBSs. You should consider the following as a check-list to make sure your WBSs do not picture this:

- This is not a WBS: it is quite typical that students creating a website map instead of a WBS when planning for a web development project. There is also a confusion between product architecture with a artifact-based WBS.
- Mix of WBS types: we recommend students to choose between artifact-based or task-based WBS, but not mixing both of them. It has been observed that students creating some branches of their WBS with a hierarchy of artifacts, while some other branches with a hierarchy of tasks. It is difficult to do time planning, task assignment when referring to this WBS.
- Violation of 100% rule: most of first attempt creating WBS by students violate the 100% rules. For instance, when thinking about the tasks they need to perform, the students

often forget about project planning and management. In fact, they spend a significant amount of time for these tasks. Or students also spend a lot of time for learning new technologies. But when they think about the structure of their products, they do not make a room to include the learning activity in the WBS. Consequently, hundreds of team working hours are not reflected in their WBS, and in their time plan.

- Violation of the mutual exclusive rule: there are WBS with overlapping nodes. For instance, one put usability testing as a part of the Testing node, as well as a part of Project deployment node.
- Right level of details: some WBSs have only one level down, while some others have four, five levels down. Not every node needs to be broken down to the same number of levels. However, the level of details should be determined to keep the balance between the overview of the project as well as the detail of the tasks.
- Numbering WBS's nodes: it is a common mistake that students do not numbering their WBS. After creation of the work breakdown structure, levels should be numbered for ease of location. For example, a project is divided into 2 major deliverables and these are numbered as 1 and 2 respectively. Then, work packages under each major deliverable are numbered as 1.1, 1.2 etc. for major deliverable 1 and 2.1, 2.2 etc. for major deliverable 2.
- Forget to update WBS: as the knowledge about the project evolves, there is a need to update your WBS to reflect correctly the current scope of the project. Maintaining an accurate version of WBS will provide a consistent reference when discussing, assigning tasks and to update time and project plans.

Tip 12: In general, it is preferable to look at task-based WBS with three levels of details

3.3 Time planning

The essential of time planning is to create and update a project schedule diagram called Gantt Chart. A Gantt chart, or harmonogram, is a type of bar chart that illustrates a project schedule[2]. This chart lists the tasks to be performed on the vertical axis, and time intervals on the horizontal axis. The width of the horizontal bars in the graph shows the duration of each activity. Gantt charts illustrate the start and finish dates of the terminal elements and summary elements of a project. Terminal elements and summary elements constitute the work breakdown structure of the project. Modern Gantt charts also show the dependency (i.e., precedence network) relationships between activities.

[2] https://en.wikipedia.org/wiki/Gantt_chart

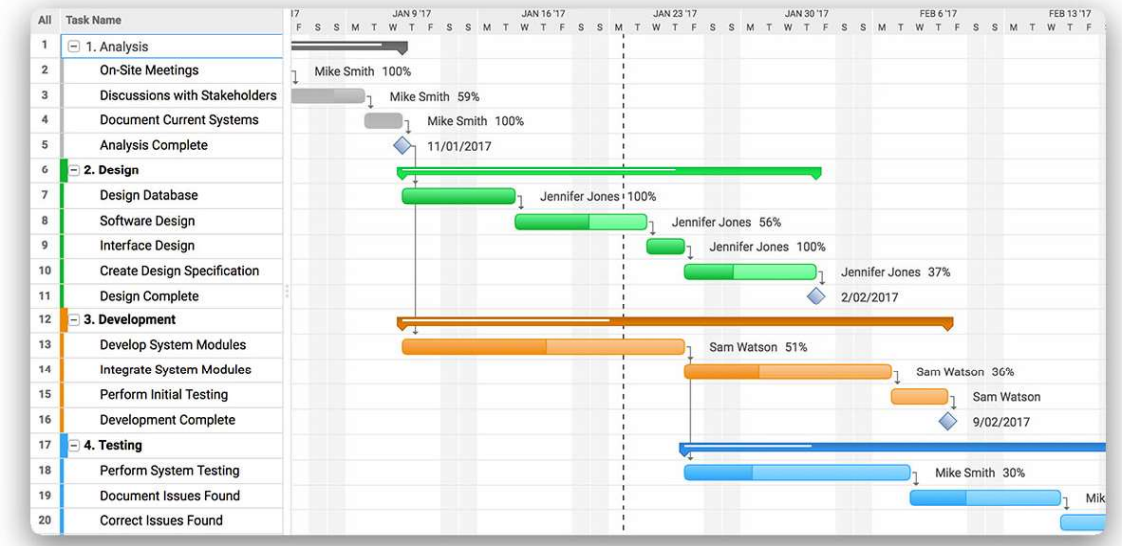

Figure 8: An example of a Gantt chart for a web application development project

Similarly to Section 3.2, we present here common mistakes when students firstly making Gantt charts:

- This is not a Gantt chart: there are common conventions about drawing Gantt chart. Students should use existing templates for visualizing their Gantt Chart, and not invent their own notations. Microsoft Office provides a template as a Excel document[3], but you can find various templates in other software online.
- Right size of Gantt chart: as seen in Figure 8, the overview of the whole project is capture in one screen. It is possible to go in details of the WBS and have the correspondingly timeline for each task. However, keep in mind that too many details can make it difficult to watch the overview. Some modern project management tools might provide features to dynamically switch between views of Gantt Chart. However, most of the free templates will not give you the capacity.
- Forget to add milestones: Gantt chart includes not only timeline for activities, but also important milestones, such as Sprint review meeting, User acceptance testing, Project final presentation, and Final exam date. Visualizing milestones will guide the adjustment of time plan to meet project deadlines.
- Forget to update Gantt chart: similar to WBS, time plan can change over time. There is also a need to update the schedule chart when WBS is updated. Maintaining an accurate version of WBS is important to have a correct view on the project progress and the time plan till the end of the project.

Tip 13: Students should use the Excel template for Gantt Chart providing in this book for their time plan

[3] https://templates.office.com/en-us/simple-gantt-chart-tm16400962